Unit V

CHAPTER 15

DATA MANIPULATION THROUGH SQL

Learning Objectives

After the completion of this chapter, the student will be able to write Python script to

- Create a table and to add new rows in the database.
- Update and Delete record in a table
- Query the table
- Write the Query in a CSV file

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15.1 Introduction

A database is an organized collection of data. The term "database" can both refer to the data themselves or to the database management system. The Database management system is a software application for the interaction between users and the databases. Users don't have to be human users. They can be other programs and applications as well. We will learn how Python program can interact as a user of an SQL database.

15.2 SQLite

SQLite is a simple relational database system, which saves its data in regular data files or even in the internal memory of the computer. It is designed to be embedded in applications, instead of using a separate database server program such as MySQLor Oracle. SQLite is fast, rigorously tested, and flexible, making it easier to work. Python has a native library for SQLite. To use SQLite,

Step 1 import sqlite3

Step 2 create a connection using connect () method and pass the name of the database File

Step 3 Set the cursor object cursor = connection. cursor ()

- Connecting to a database in step2 means passing the name of the database to be accessed. If the database already exists the connection will open the same. Otherwise, Python will open a new database file with the specified name.

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